SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Section	:	3	

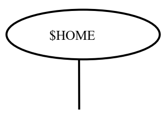
Instruction: Please answer all of the following questions. Whenever the ✋ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo “Hello world” > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo “Honda Accord” > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

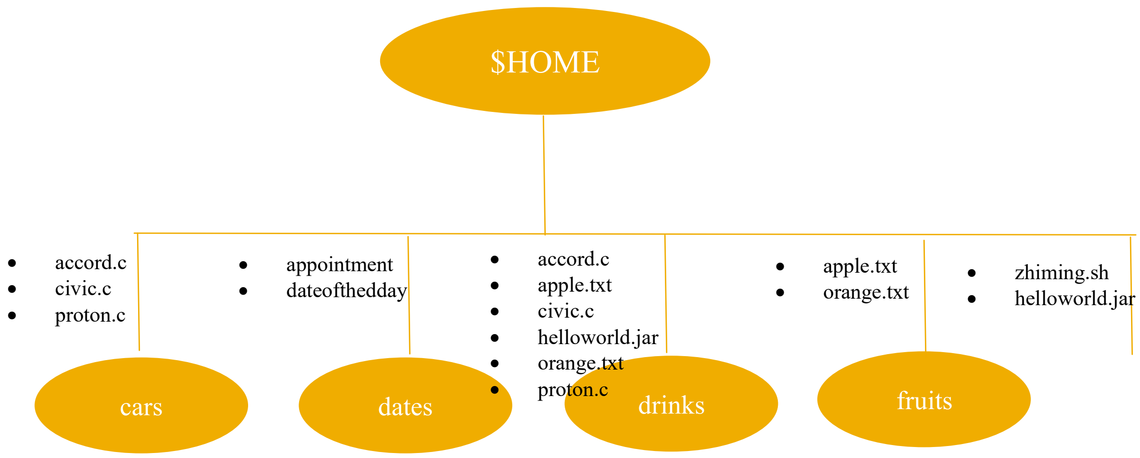
[4 marks]



Print screen the script that you type;

Then draw the tree

```
zhiming@secr2043:~$ chmod u+x zhiming.sh
zhiming@secr2043:~$ ./zhiming.sh
zhiming@secr2043:~$ tree
.
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── helloworld.jar
└── zhiming.sh
```



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
zhiming@secr2043:~$ chmod u+x interactivefiles.sh
zhiming@secr2043:~$ ./interactivefiles.sh
Enter the file extension c
Files with .c extenion:
./workspace/multiply.c
./workspace/hello_world.c
./workspace/kernel_info.c
./workspace/argument.c
./workspace/hello_u.c
./workspace/input_keyboard.c
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 12
!/bin/bash

ead -p "Enter the file extension " ext
cho "Files with .${ext} extenion:"
ind . -type f -name "*.${ext}"

ount=$(find . -type f -name "*.${ext}" | wc -l)
cho "Number of files with .${ext} extension: $count"
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

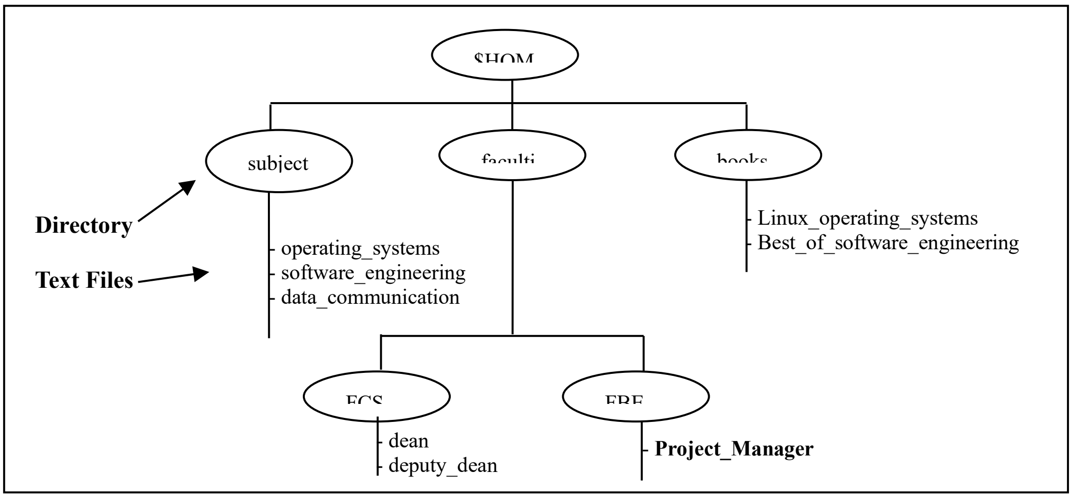
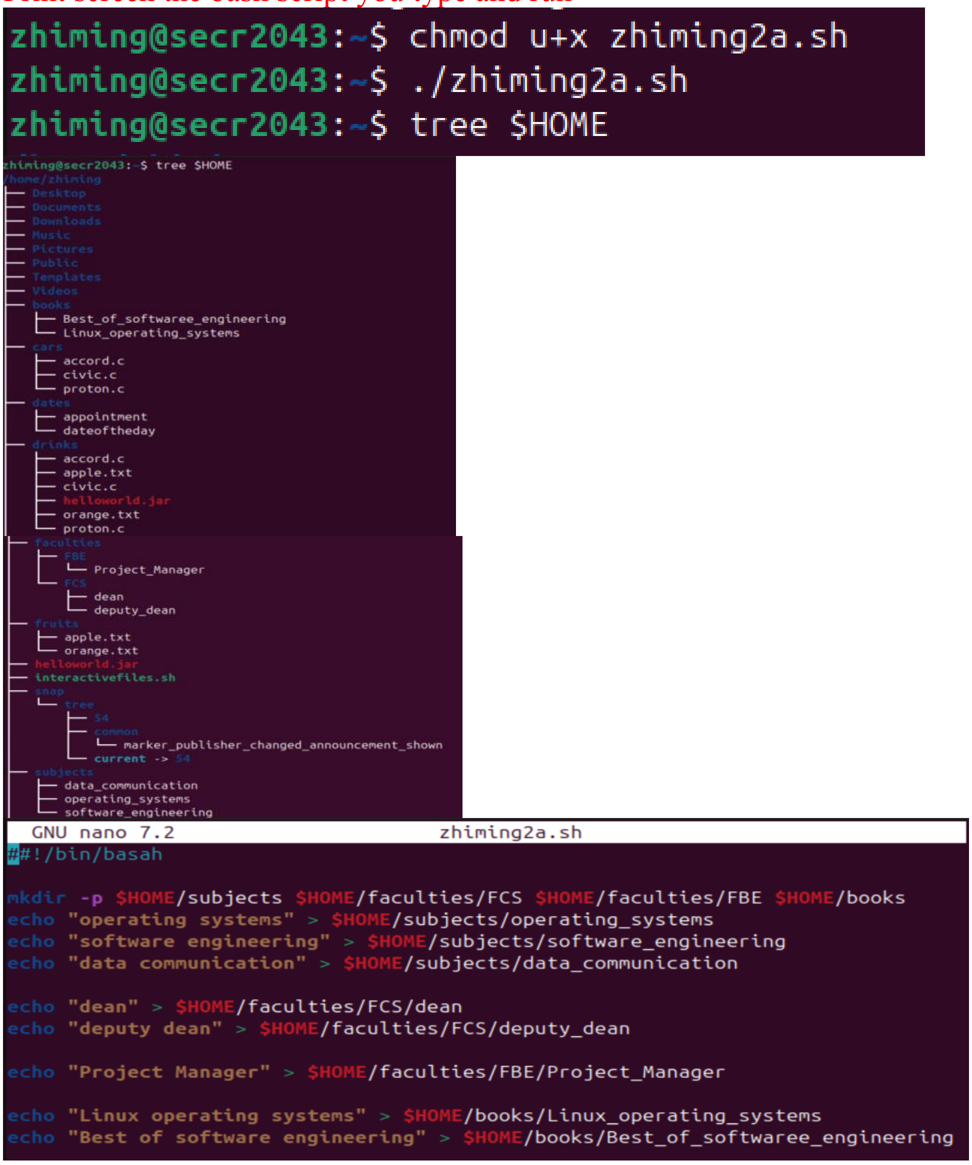


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run



b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
`books`	`drwxrwxr-x`
`subjects`	`drwxrxr-x`
`Best_of_software_engineering`	`-rw-rw-r--`
`FCS`	`drwxrwxr-x`
`project_manager`	`-rw-rw-r--`

```
drwxrwxr-x 2 zhiming zhiming 4096 Jun 19 11:55 books
drwxrwxr-x 2 zhiming zhiming 4096 Jun 19 11:55 subjects
-rw-rw-r-- 1 zhiming zhiming 29 Jun 19 13:50 Best_of_softwaree_engineering
drwxrwxr-x 2 zhiming zhiming 4096 Jun 19 11:55 FCS
-rw-rw-r-- 1 zhiming zhiming 16 Jun 19 13:50 Project Manager
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
`subjects`	✓	✓	✓	✓	x	x	✓	x	x
`Best_of_software_engineering`	✓	x	✓	x	✓	x	x	x	x
`FCS`	✓	✓	x	x	x	x	✓	✓	✓
`project_manager`	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
  GNU nano 7.2                                    zhiming2c.sh
#!/bin/bash

chmod 744 $HOME/subjects
chmod 520 $HOME/books/Best_of_softwaree_engineering
chmod 607 $HOME/faculties/FCS
chmod 031 $HOME/faculties/FBE/Project_Manager
```

```
zhiming@secr2043:~$ chmod u+x zhiming2c.sh
zhiming@secr2043:~$ ./zhiming2c.sh
chmod: cannot access '/home/zhiming/books/Best_of_softwa
 file or directory
zhiming@secr2043:~$ nano zhiming2c.sh
zhiming@secr2043:~$ ./zhiming2c.sh
 drwxr--r-- 2 zhiming zhiming 4096 Jun 19 11:55 subjects
-r-x-w---- 1 zhiming zhiming 29 Jun 19 13:50 Best_of_softwaree_engineering
drw----rwx 2 zhiming zhiming 4096 Jun 19 11:55 FCS
 -----wx--x 1 zhiming zhiming 16 Jun 19 13:50 Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
`subjects`	`drwxr--r--`
`Best_of_software_engineering`	`-r-x-w----`
`FCS`	`drw----rwx`
`project_manager`	`-----wx--x`

End of Lab 3

**** All the Best for Final Exam ****